

Topic

- What is Multiclass Classification
 - How Logistic Regression handles multiple classes
 - One-vs-Rest (OvR) concept
 - Real-life examples
 - Practical implementation
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1. Learning Objectives

- Understand **multiclass classification**
 - Differentiate **binary vs multiclass**
 - Learn how Logistic Regression handles multiple classes
 - Understand **One-vs-Rest strategy**
 - Build a multiclass classification model
 - Interpret predictions and probabilities
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2. Quick Recap (Connect Previous Session)

“Till now we worked with binary classification where output had only 2 classes. But in real life, many problems have more than 2 categories. Today we will learn how to handle such problems.”

3. What is Multiclass Classification?

3.1 Definition

Multiclass classification is a type of classification where the target variable has **more than two classes**.

3.2 Simple Meaning (Most Important Line)

“When output has more than two categories, it is called multiclass classification.”

3.3 Examples

Problem	Classes
Student Grade	A, B, C
Animal Type	Dog, Cat, Cow
Iris Flower	Setosa, Versicolor, Virginica

4. Binary vs Multiclass

Feature	Binary	Multiclass
Classes	2	More than 2
Example	Pass/Fail	Grade A/B/C
Output	0 or 1	0,1,2...

Teaching Line

“Binary is a special case of multiclass classification.”

5. How Logistic Regression Handles Multiclass?

Key Idea

Logistic Regression is originally for binary classification.

👉 To handle multiclass, we use:

👉 **One-vs-Rest (OvR)**

6. One-vs-Rest (OvR) Strategy

Concept

For each class:

- Treat that class as **1**
 - Treat all other classes as **0**
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Example (3 classes: A, B, C)

We create 3 models:

1. A vs (B + C)
 2. B vs (A + C)
 3. C vs (A + B)
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Final Prediction

- Model calculates probability for each class
 - Highest probability wins
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“In multiclass classification, the model chooses the class with the highest probability.”

7. 📊 Probability Output

Instead of single probability:

👉 Model gives:

Class A	Class B	Class C
0.2	0.7	0.1

👉 Final output:

Class B (highest probability)

8. 📊 Real-Life Example

Example: Student Grade Prediction

Marks	Grade
40	C
70	B
90	A

👉 Model learns:

- Low marks → C
 - Medium → B
 - High → A
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9. 📊 Decision Boundary

Binary:

- One boundary

Multiclass:

- Multiple boundaries

👉 Space divided into multiple regions

10. 📊 Key Points

- ✅ Works for more than 2 classes
- ✅ Uses probability for each class

- ✓ Chooses highest probability
 - ✓ Uses OvR internally
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11. 🎯 Interview Concepts

Q1: What is multiclass classification?

👉 Classification with more than 2 classes.

Q2: Can Logistic Regression handle multiclass?

👉 Yes, using One-vs-Rest.

Q3: What is OvR?

👉 One class vs all other classes.

Q4: How prediction is made?

👉 Class with highest probability is selected.

12. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Predict student grade:

- A, B, C

Multiclass classification extends binary classification to handle multiple categories.

Logistic Regression uses One-vs-Rest strategy and selects the class with the highest probability.

This makes it useful for real-world problems like grading, classification, and categorization.”